Low-Light Image Enhancement with Cross-Dataset Generalization

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Abstract

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1. Introduction

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2. Experimental Protocol

2.1. Datasets

The pipeline developed in this work is based on four datasets, each with a specific role in the experimental protocol: (i) LOL-v2 [4], (ii) LOL-v1 [3], (iii) MILLs [2], and (iv) ExDark [1]. All models are trained on a single source domain and are then evaluated on additional datasets without fine-tuning. This choice is intended to measure cross-dataset generalization, i.e., the ability of each model to enhance low-light images whose distribution differs from the training data.

The main training dataset is LOL-v2. This dataset contains both a real-captured subset, hereafter referred to as LOL-v2 Real, and a synthetic subset, referred to as LOL-v2 Synthetic. In the main protocol, only LOL-v2 Real is used, meaning that the low-light/normal-light pairs are acquired from real scenes rather than obtained through synthetic low-light degradation. LOL-v2 is a paired dataset: each low-light image is associated with a corresponding normally exposed reference image, which can be used as the target for supervised training and evaluation. The official training split of the real-captured subset is used to train all models, while the official real-captured test split is kept for in-domain evaluation. In this context, "in-domain" means that training and testing images come from the same dataset and acquisition setting. The synthetic subset is kept separate from the main protocol and is used for an additional ablation study, aimed at evaluating whether synthetically degraded low-light examples improve or degrade generalization.

LOL-v1 is used only as a paired cross-dataset evaluation set. It represents a relatively close cross-dataset scenario, because it belongs to the same general benchmark family as LOL-v2.

MILLs is used as a second paired cross-dataset evaluation set. In particular, the low-resolution DSLR split, MILLs-DSLR, is adopted to keep the evaluation computationally manageable. Unlike LOL-v1, MILL introduces a controlled illumination setting. Its input images are associated with different illumination levels, and each input has a corresponding well-illuminated reference image. This makes it possible to evaluate the models not only with global results over the whole dataset, but also by studying how performance changes as the input illumination becomes more severe.

Finally, ExDark is used as an unpaired stress-test dataset. Here, unpaired means that the dataset provides low-light images without a corresponding normally exposed reference image. Therefore, ExDark is not used for reference-based comparison, but for no-reference evaluation and qualitative failure analysis on real-world dark scenes. Its category-based organization is preserved, meaning that images remain grouped according to the object category provided by the dataset, such as cars, people, animals, or indoor objects. This organization may help inspect whether some failure cases are related to the type of scene or object.

2.2. End-to-End Pipeline

The experimental workflow is organized as a sequence of modular stages, in which each stage performs a specific task and produces outputs for the following stages, from the raw datasets to the final quantitative and qualitative evaluation. The planned pipeline consists of (i) manifest generation, (ii) offline preprocessing, (iii) model training, and (iv) model evaluation.

The only dataset-specific stage is manifest generation. Besides its natural role of organizing the data, this step is intentionally designed to convert datasets with different structures into a common format. As a result, the subsequent stages of the pipeline can operate on the same data representation without relying on the original organization of each dataset, making them effectively dataset-agnostic.

2.2.1. Manifest Generation

The purpose of the first stage of the pipeline is to scan the original dataset folders and build CSV manifest files from their structure. In each manifest, every row represents one image sample and stores the dataset name, the split to which the sample belongs, the path of the low-light input image, the path of the corresponding target image when available, and optional metadata such as the image category or the illumination level.

This step is necessary because the datasets used in this project are not organized in the same way. Some datasets provide paired low-light and normal-light images with the same filename, while others use different filename conventions. ExDark is instead unpaired and organized by object category. MILLs also includes an illumination level in the input filename, which is useful for analyzing how performance changes under different lighting conditions. Without this standardization step, the subsequent scripts in the pipeline would need to contain several dataset-specific rules.

For this reason, manifest generation is the only intentionally dataset-specific part of the pipeline. The script reads a YAML configuration file that describes where each dataset is located, which output file should be produced, and how the corresponding manifest should be created. In LOL-v1 and LOL-v2 Synthetic, low-light images and their corresponding reference images share the same filename, making the association straightforward. In LOL-v2 Real, the filenames differ only because low-light and reference images use different prefixes. ExDark follows a completely different structure: images are grouped by category and no reference image is provided, so only the image path and category information are recorded. Finally, in MILLs, the filename also contains information about the illumination level, which is extracted and stored together with the path of the corresponding reference image.

The output of this step is a CSV manifest file for each dataset, with a fixed column structure. This common format is what makes the later stages dataset-agnostic: once the manifests have been created, preprocessing, training, and evaluation only need to read the paths and metadata stored in the CSV files, without knowing how the original dataset was arranged on disk.

The manifest generation script also supports reproducibility-oriented choices. When a dataset does not provide an explicit validation set, a validation split can be sampled from the training split using a fixed random seed. Moreover, the generated rows are sorted according to the split and image path, making the row order deterministic across runs.

3. Conclusion

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References

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